eBnC Onboarding for Balance and Control on DAM Delta events processing

Introduction

We need to put automated controls to ensure no data loss (missing/duplicate change events) between DAM and CRR when it comes to consuming DAM delta change events by CRR to comply with TECH06.48 Global Automated Controls and Balancing Standard. For this, we will be using the Enterprise Balance and Controls (eBnC) solution.

EBNC broadly offers two types of balancing.

1. **ID/Batch based balancing** - For data flows/systems where source and destination communicate in batches. Balancing is performed in real time whenever destination asks to balance.
2. **Window based balancing** - For data flows/systems where source and destination communicate in real time. Balancing is performed on a pre-configured time windows.

Note: eBnC is not limited to above two use cases only. They can also provide support for systems where source is sending data in real-time but the destination is consuming the same in batches. Application teams can consult their usecase with the eBnC onboarding team to understand if eBnC can support their usecase or not.

We are choosing window based balancing for below two reasons.

1. It eliminates the need to build logic between source and destination to define end of messages and batching.
2. It would be sacalable when DAM (source) starts sending delta events to CRR (destination) multiple times a day.
3. Auto-rebalancing of late events.

Refer following link to know more on eBNC What is eBnC

Quick Help

Reach out to the eB&C team: eabncdev@aexp.com

Slack #ea-bnc-infoshare

For details on how to integrate with eBnC: eBnC Training

To engage eBnC, please submit a request at: *https://ebncui.aexp.com/BNCEngagements*

Onboarding Process

Below form needs to be filled with required details of source and destination and successful submission will produce an engagement id as reference tracking number.

https://ebncui.aexp.com/BNCEngagements

Schedule a discussion with eBnC team to explain your usecase to get recommendation on right configuration and make changes in the engagement details if required.

After completing the final details ask the team to setup the configurations in E1 environment.

Integration Steps

Publisher endpoint E1 for Source and Destination
A2A - https://ebnc-dev.aexp.com/bnc/eventpublisher
B2B/EAG - https://eag-dev.aexp.com/ebnc/operations/v2/events/publisher

Balancer endpoint E1 for Destination (Not required for Window based Balancing)
A2A - https://ebnc-dev.aexp.com/bnc/eventbalancer
B2B/EAG - https://eag-dev.aexp.com/ebnc/operations/v2/events/balancer

Ebnc E1 UI : https://ebncui-dev.aexp.com/

IDaasToken API
A2A - https://oneidentityapi-dev.aexp.com:443/security/digital/v1/application/token
B2B/EAG - https://eag-dev.aexp.com/oneidentity/security/digital/v1/application/token

Please follow these steps in order to send events to eBnC and to view the eBnC ui.

1. You will need to go to Idaas portal and request a connection to ebnc (aim id 600000548): https://idaas.aexp.com/idaas/

   Raise access to api's under **ebncprovider** and **eBnCEAGProvider**: https://enterprise-confluence.aexp.com/confluence/display/BnC/eBnC+-+IdaaS+Requests

   /bnc/eventpublisher. POST
   /ebnc/operations/v2/events/publisher POST
2. Once that is done you will need to create an authentication token and use it in the REST calls to EBnC: https://enterprise-confluence.aexp.com/confluence/display/OIST/App-to-App+Security
   a. Refer the Postman Collection section for quick setup.
   b. For more details here is quick access to jwt token api guide: A2A Consumer Integration Guide
3. To create a request for an access to EBnC UI: https://enterprise-confluence.aexp.com/confluence/display/BnC/How+to+Request+eBnC+UI+Access
4. Swagger url: https://ebnc-dev.aexp.com/swagger-ui/index.html#

Postman Collection

Import collection and environment jsons in postman and replace the value of **secret** variable in the environment with the idaas app id actual secret for that environment and you are set to test from local.

A2A ebnc.postman_collection.json

A2A E1.postman_environment.json

EAG ebnc.postman_collection.json

EAG E1.postman_environment.json

Onboarding Details

eBnC Engagement Id	121b5d0a-a330-49cb-b21e-3c776a07c8da
IDaaS Integration Id - CRR	e7eb53bf-da1c-3fea-8c37-b83aa5d1cc74
IDaaS Integration Id Owners	Richa Gupta Nishant Kashyap Ranjana Panwar Gopesh X Verma
CRR Ads Group who can see the eBnC flows created for CRR (600003728) on eBnC UI	GG-EBNC-APPLICATION-READER-600003728
Onboarding Status	Done
E1 Configuration Status	Configuration Done
E1 Basic Validation & Testing via Postman	▪ Validation - Able to publish events to eBnC via EAG and A2A publisher endpoints ▪ Testing - Yet to test
E1 Integrated Testing Status	
E2 Configuration Status	

E2 Basic Validation & Testing via Postman	
E2 Integrated Testing Status	
E3 Configuration Status	
E3 Basic Validation & Testing via Postman	
E3 Validation Status	

Guidelines for E1 to E2 and E2 to E3 Migrations

Please follow the below guidelines for the Production Configuration Setup Requirements.

E2 Configuration Setup Requirements

1. There is a 1day lead time for migrating configuration from e1 to e2 environment.
2. Email communication to eabncdev@aexp.com with config details FlowId, SourceEventName and DestinationEventname

E3 Configuration Setup Requirements

1. There is a lead time of 3 business days for migrating configuration to production.
2. Email communication to eabncdev@aexp.com with config details FlowId, SourceEventName and DestinationEventname

E2 Sign-off

1. They need our testing results signed off from our business and/or GFII.

E2 Testing

1. Testing should cover positive and negative scenarios and should be evident through the UI for validation
   Real-time / Window balancing
   - Positive – Valid events and successful balancing
   - Negative – No source event found
   - Negative – No destination event found
   - Negative – Both events found but amounts/counts don't match

1. Business partners(or) GFII/our team should have set up access to the UI

eBnC Production sign-off requirements

They expect that we are aligned with the following before proceeding to E3:

1. All events are immutable and have a durable UUID/Rowkey and a Unique Timestamp associated with the event
2. Any retry of events should have the same timestamp
3. Application should have re-try mechanism for calling API's
4. If same events are expected to be reprocessed deliberately – then we can leverage the Version number to indicate so (V2 instead of V1 and so on…)
5. All production incidents are raised against the destination system as per the configuration and we confirm that production support can handle the incidents
6. eBnC will not be able to action on Incidents to get Out of Balance / UNBALANCED (OOB) information through ServiceNow
7. We will need to leverage the UI or the /bnc/status API endpoint for automation and OOB analysis
8. Any subsequent changes to the configuration require lead time and will be scheduled based on pre-determined schedule